Sahaj Bhadja

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Summary

Software Engineer with 2+ years delivering **business application enhancements**, **SQL-backed services**, and **test-driven defect fixes** using Java/Python/JavaScript in Agile, stakeholder-facing environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Implemented **low-to-medium complexity application components** on **Azure SQL/PostgreSQL** for 500+ users, reducing latency from **1.2s to 350ms** through schema tuning and query optimization.
- Developed **unit/system validation workflows** for 5K+ monthly jobs, improving defect detection, documentation quality, and readiness for production deployment with cross-team stakeholders.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Built and tested **Flask + React features** supporting 10K+ daily requests, translating business requirements into maintainable APIs and user-facing components under Agile project methodology.
- Improved **delivery predictability** through CI/CD controls and concurrent-processing checks, reducing turnaround time by **60%** while maintaining stable production behavior and defect resolution speed.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Developed **Python testing pipelines** over 100K+ trajectory logs, supporting reproducible analysis, system-level debugging, and clear communication of findings during design and implementation reviews.
- Created technical reports and recommendations from test outcomes, helping teams prioritize fixes and refine feature behavior with lower risk of regressions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Implemented **API and ETL services** using OOP, AWS Glue, and S3, improving response times by **30%** and stabilizing application-level data exchange across systems.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito across three OS environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built **REST API and WebSocket modules** for eight workflows and 2M+ synchronized records, improving maintainability and reliability of department-level business application processes.
- Automated testing and deployment using **Docker** and Jenkins, reducing deployment time by **50%** and increasing confidence in repeatable production releases.

Skills

Languages: **Java**, **JavaScript/TypeScript**, **Python**, SQL, C++
Business App Development: SDLC, Agile/Waterfall familiarity, requirements capture support, low/medium complexity implementation
Data & Storage: SQL, Oracle familiarity, PostgreSQL, MySQL, MongoDB, data modeling and manipulation fundamentals
Web/API: React.js, Flask, FastAPI, Node.js, REST APIs, HTML, XML
Quality & Delivery: unit/system testing, test plans and cases, Git/GitHub, CI/CD, defect triage, stakeholder communication

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built a **department-level SQL application tool** across 10+ databases with schema-aware query generation and execution validation, improving quality and speed of internal reporting workflows.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Developed **business workflow modules** supporting 50K+ users and 1M+ transactions with secure APIs and scalable architecture, demonstrating strong customer-facing solution quality.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Implemented **application support and monitoring workflows** for 100+ daily appointments, improving record retrieval by **2x** and reducing issue resolution time in production.